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Paper Code: TBT-101

(B. Tech Biotechnology)
End Semester Examination, December 2024
I Semester

Paper Name: FUNDAMENTALS OF MEDICAL MICROBIOLOGY

Time: 3:00 Hours

MM: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any TWO sub questions among a, b & c in each main question.
- (iii) Total marks for each main question are twenty.

Q1.

- (a) Define infection and disease? With examples, describe the different modes of transmission of infectious agents. How do vectors, such as mosquitoes and ticks, act as intermediaries in the transmission of infectious agents? CO1
- (b) Highlight the natural microbiota of skin? How does the skin act as a barrier against infection, and what microbial pathogens can compromise its role as a portal of entry for pathogens? CO1
- (c) Comment on the general morphology, source of infection of *Clostridium tetani*? Differentiate the types of tetanus caused by *C. tetani*. CO1

Q2

- (a.) Describe the general morphology of fungal pathogens responsible for causing mycoses. How do yeasts, molds, and dimorphic fungi differ in terms of structure and growth? CO2
- (b.) Describe the morphology of Rhabdovirus? Using diagram, explain the pathogenesis and laboratory diagnosis of Rabies infection? CO2
- (c.) Explain how ELISA is applied in the detection of Hepatitis B virus (HBV) infection? What factors influence the sensitivity and specificity of an ELISA test in diagnosing viral infections CO2

Q3.

- a. How is the hemagglutination test used in blood typing. Discuss the significance of this technique in detection of bacterial infection? CO3
- b. Describe the properties of immunogens that are responsible an active immune response. Elaborate the key mechanism by which acquired immune system responds to a bacterial infection? CO3
- c. Describe the structure of an immunoglobulin and their different classes. How does its structure relate to its function? CO3

Q4.

- (a) Describe the significance of selective and differential media in Medical microbiology. How Mannitol Salt agar and Blood agar are useful in the differentiation of pathogenic bacteria in clinical laboratories? CO4
- (b) Discuss the main methods of antibiotic susceptibility testing, including the disk diffusion method (Kirby-Bauer), broth dilution method? CO4
- (c) How does the National Program for Control of Vector-Borne Diseases address the surveillance and monitoring of vector populations? Elaborate with suitable examples. CO4

Q5.

- (a) Differentiate between attenuated and inactivated vaccines? Discuss on the types of vaccines available for the control of viral infection. CO5
- (b) Discuss the role of genetic engineering and basic biomedical tools used in the treatment of cancer? CO5
- (c) Describe the significance of Hybridoma technology in production of monoclonal antibodies. Also, discuss the application of monoclonal antibodies in Medical Biotechnology. CO5